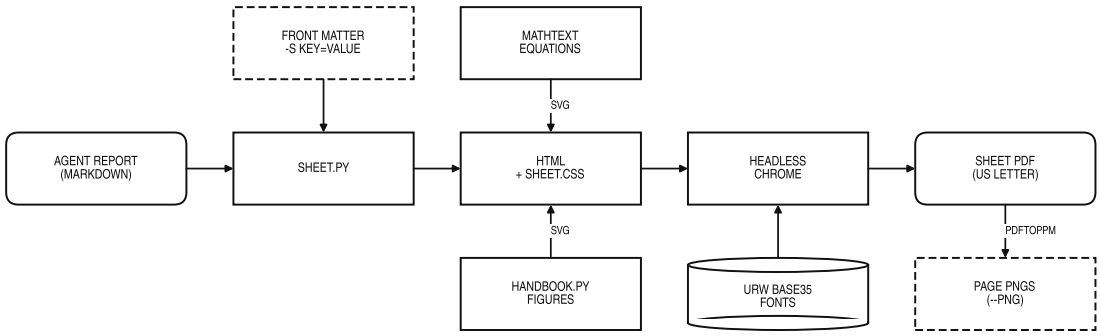
APOLLO STYLE REVIEW SHEETS

Reference and specimen sheet. Every block the format has, with its source beside it. Start a new sheet from template.md.

A.1 FUNCTIONAL DESCRIPTION.

A review sheet is a section of a handbook made from one agent's report: numbered paragraphs, figures and tables with drawing numbers, call-outs, checklists and equations, printed in black on US Letter to be read with a pen. The report is plain Markdown; `sheet.py` renders it as it is, and the `:::` blocks below add the handbook furniture. Figures are drawn by `handbook.py` as line art with live lettering, equations by matplotlib's mathtext, and the page by headless Chrome from HTML and `sheet.css`.



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Figure A-1. Review Sheet Functional Diagram

```
python sheet.py report.md                          writes report.pdf beside it
python sheet.py report.md -o out.pdf --png         and page PNGs, to look without a printer
python sheet.py notes.md -s section=3 -s tab=PHY   someone else's report, without editing it
```

A.2 PAGE DESIGN.

Table A-1. Design Rules

Element	Set in	Rule
Body	Nimbus Roman 9.6 pt on 1.32	justified and hyphenated; the first line after a heading indented 0.3 in
Lettering on figures	Nimbus Sans Narrow, 5 to 7 pt	capitals only; the same face live in the PDF, never outlines
Checklists, code	Nimbus Mono PS 8.3 pt	a checklist is boxed, under a black bar carrying its title and role
Page	US Letter	margins 0.95 in top, 0.85 bottom, 0.45 left, 0.3 right; a 0.95 in gutter for numbers
Ink	black only	white paper to print; `paper: cream` for a screen copy
Headings	## capitals, ### underlined	the paragraph number hangs in the gutter: A.4, A.4.2
Numbers	Figure A-1, Table A-1, (A-1)	figures and tables count through the section; table titles above, captions below
Drawing numbers	APO-501A ...	one per figure, bottom right, from the tab
Strokes	0.6 to 0.9 pt	dashed for optional or planned, hatched for used or selected, solid black for on
Series	dash pattern, end label	solid, dashed, dotted, dash-dot; labelled at the line's end, never by colour or legend
Header, footer	doc, handbook, system; mission, dates, page	dates MM.DD.YYYY; page A-n from the section
Changes	bar in the gutter	`:::changed` beside anything altered after the basic date; set `change:`

The page is quiet on purpose: no colour, no shading but hatch, no rules but the table frames and the `---` line. Anything that needs colour to be read needs redrawing, not colouring; a colour screen capture prints inverted to ink with `gray=1`.

A.3 TEXT AND CALL-OUTS.

A.3.1 Headings and paragraphs.

A `#` heading names the sheet when there is no `title`; a later `#` starts a part on a new page. `##`, `###` and `####` are numbered from `section`, and any number the report gave its own headings is dropped. Inline marks: `**bold**`, `*italic*`, `_underline_`, backticks for code, `^sup^`, `~sub~`, `$math$`, and `--` for a dash. Links keep their words; paper has no links. A line starting `<!--` opens a comment that never prints, up to `-->`.

A.3.2 Call-outs and checklists.


```
:::caution
Text that, if ignored, costs
time or data.
:::

:::warning
Text that, if ignored, damages
hardware or loses work for good.
:::

:::note
Anything worth a box that is
neither.
:::

:::checklist Render a Sheet | Any
1  python sheet.py r.md --png
2  read every page PNG
3  fix, render again
:::

:::changed
A paragraph altered after the
basic date.
:::
```

CAUTION

Text that, if ignored, costs time or data.

WARNING

Text that, if ignored, damages hardware or loses work for good.

NOTE

Anything worth a box that is neither.

RENDER A SHEET	ANY
1	python sheet.py r.md --png
2	read every page PNG
3	fix, render again

A paragraph altered after the basic date. The bar is in the gutter, beside the paragraph number.

A.4 MATHEMATICS.

A.4.1 Inline and display.

Inline math sits on the baseline of the text: the corner of an RC low-pass is $f_c = 1/(2\pi RC)$, written `$f_c = 1/(2\pi R C)$`. A display, `$$ ... $$` on its own lines, is centred and numbered from the section:

$$|H(f)| = \frac{1}{\sqrt{1 + (f/f_c)^{2n}}} \tag{A-1}$$

where f = frequency *Hz*
f_c = corner frequency, where the gain is down 3 dB *Hz*
n = order of the Butterworth filter

The `:::symbols` block above is the handbook's "where" list: one `SYMBOL | MEANING | UNIT` a line, the symbol in TeX without dollars.

A.4.2 Aligned equations.

`:::equations` sets one equation a line, aligned on its first relation (or on `&` where a line has one). A line that starts with a relation continues the one above and takes no number. Here, the length of an audio block:

```
:::equations
T_{\mathrm{block}} = N / f_s
= 128 / 44100\ \mathrm{s}
\approx 2.902\ \mathrm{ms}
G_{\mathrm{dB}}(f_c) = {-10} \log_{10} 2
\approx {-3.01}\ \mathrm{dB}
:::
```

$$T_{\mathrm{block}} = N/f_s = 128/44100\ \mathrm{s} \approx 2.902\ \mathrm{ms} \quad \text{(A-2)}$$

$$G_{\mathrm{dB}}(f_c) = -10\log_{10} 2 \approx -3.01\ \mathrm{dB} \quad \text{(A-3)}$$

A.4.3 A result, boxed.

`$$\boxed{...}$$` frames the line a reader should carry away:

$$\boxed{f_c = \frac{1}{2\pi RC}} \quad \text{(A-4)}$$

CAUTION

Equations are set by matplotlib's mathtext, a subset of TeX: fractions, roots, sub- and superscripts, Greek, `\left( \right)`, accents, `\mathrm`, `\mathbf`, `\mathcal`, sums and integrals. It has no environments (`\begin{aligned}`, matrices, `cases`): use `:::equations` for alignment, a table for a matrix. `\text{}` is rewritten to `\mathrm{}` and `\sqrt x` to `\sqrt{x}`. What will not parse prints as its TeX in typewriter, so nothing is lost: `\begin{pmatrix} a & b \end{pmatrix}`.

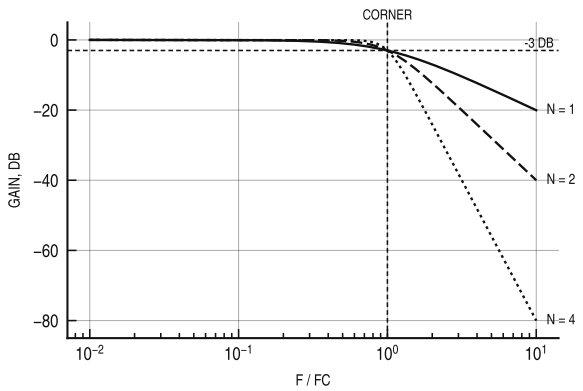
A.5 FIGURES.

Every figure is line art with a caption below and a drawing number at its lower right. `:::diagram`, `:::timeline`, `:::sequence`, `:::signals` and `:::budget` run the width of the column; a `:::plot` is 3.4 by 2.2 in unless it says `size wide` or `size W H`, so two sit side by side in a `::::row`.

A.5.1 Plots: exact functions and tabulated series.

`fn "EXPR"` draws a curve exactly from a numpy expression in `x` (`sin exp sqrt log10 pi where clip` and the like), over `range LO HI`. `y COLUMN` draws a column of a CSV against `x COLUMN`; `dots` sets it as points, the way measurements are told from a model. Thresholds are `hline`, events `mark`.

```
:::plot Butterworth Gain re the Corner
range 0.01 10
fn "-10*log10(1+x**2)" "N = 1"
fn "-10*log10(1+x**4)" "N = 2"
fn "-10*log10(1+x**8)" "N = 4"
log x
xlabel "f / fc"
ylabel "gain, dB"
ylim -85 5
hline -3.01 "-3 DB"
mark 1 "CORNER"
:::
```

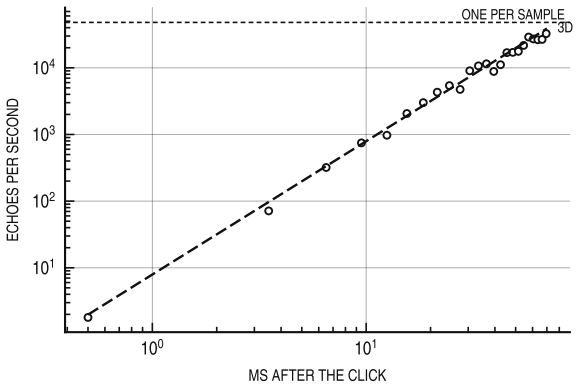


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Figure A-2. Butterworth Gain re the Corner

```
:::plot Echo Density, 4 m Cube
csv density.csv
x ms "ms after the click"
y d3 "" dots every=6
fn "4*pi*343**3*(x/1000)**2/4**3" "3D"
log xy
ylabel "echoes per second"
hline 48000 "ONE PER SAMPLE"
:::
```

The dots are a CSV of counts; the line is $4\pi c^3 t^2/V$ for c = 343 m/s and V = 64 m^3, drawn over the CSV's own x.



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Figure A-3. Echo Density, 4 m Cube: CSV (Dots) and Formula (Line)

APO

A.5.2 Block diagrams.

Figure A-1 is a `:::diagram`. Nodes sit on a grid of columns and rows (`grid DX DY` in inches, if the default 1.55 by 0.78 crowds the lettering); edges route themselves straight or with one elbow and are labelled on their longest run. Shapes: `box`, `round` for ends, `dash` for the optional, `drum` for storage, `text` for bare lettering; `fence "LABEL" ids...` draws a dashed subsystem boundary.

```
:::diagram Review Sheet Functional Diagram
grid 1.45 0.8
node fm "FRONT MATTER\n-s KEY=VALUE" 1 0 dash
node rep "AGENT REPORT\n(MARKDOWN)" 0 1 round
node sheet "SHEET.PY" 1 1
node fonts "URW BASE35\nFONTS" 3 2 drum
edge rep sheet
edge mt html "SVG"
edge pdf png "PDFTOPPM"
...
:::
```

A.5.3 Timelines.

Rows of bars on one axis, `hatch`, `open` or `solid`; `mark` for a moment across every row, `call` for a leader to lettering below the axis. Apollo 11, from launch:

```
:::timeline Apollo 11, Ground Elapsed Time
axis "ground elapsed time, hours" 0 200 ticks=25
bar "Earth orbit" 0 2.74 solid
bar "Translunar coast" 2.74 75.83 open
bar "Lunar orbit (CSM)" 75.83 135.40 hatch
bar "LM on the surface" 102.76 124.37 hatch
bar "EVA" 109.13 111.65 solid
bar "Transearth coast" 135.40 195.31 open
mark 102.76 "LANDING" dash
mark 195.31 "SPLASHDOWN" dash
:::
```

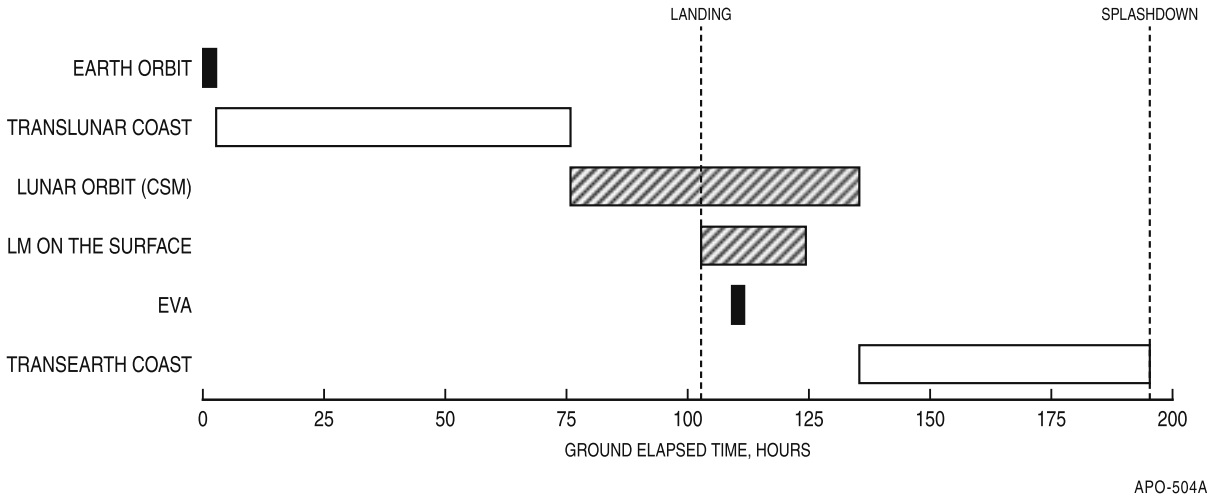
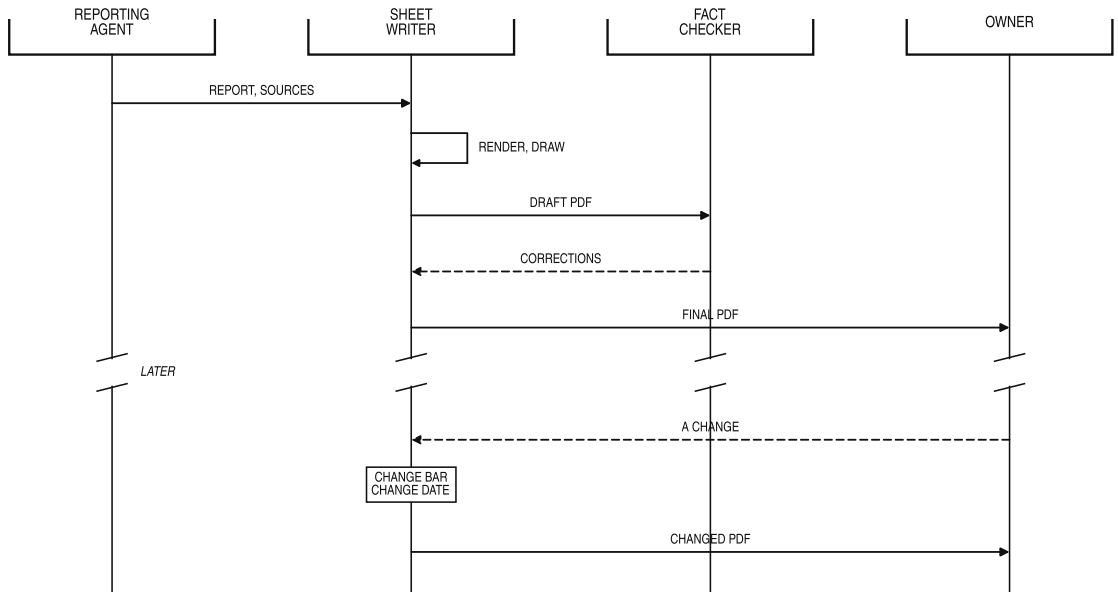


Figure A-4. Apollo 11, Ground Elapsed Time (Launch 07.16.1969, 13:32 UTC)

A.5.4 Sequence charts.

Messages between actors, one step a line, top to bottom: protocols, install flows, a review's hand-offs. `dash` for a reply, the same actor twice for its own work, `gap` where time passes. A review's hand-offs:

```
:::sequence A Review's Hand-offs
actor agent "REPORTING\nAGENT"
actor writer "SHEET\nWRITER"
actor check "FACT\nCHECKER"
actor owner "OWNER"
msg agent writer "REPORT, SOURCES"
msg writer writer "RENDER, DRAW"
msg writer check "DRAFT PDF"
msg check writer "CORRECTIONS" dash
msg writer owner "FINAL PDF"
gap "LATER"
msg owner writer "A CHANGE" dash
note writer "CHANGE BAR\nCHANGE DATE"
msg writer owner "CHANGED PDF"
:::
```



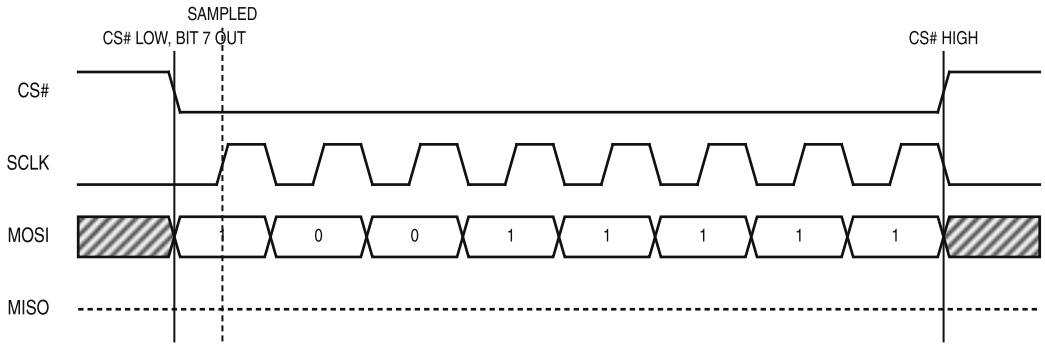
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Figure A-5. A Review's Hand-offs

A.5.5 Timing diagrams.

`:::signals` draws lines and buses on one step axis, a character a step: `0 1` levels, `p n` a clock period starting high or low, `.` hold, `z` released, `x` unknown, `=` a bus value lettered from a list. An SPI flash taking the JEDEC read-ID command 0x9F in mode 0, shifted on the falling edge and sampled on the rising:

```
:::signals SPI Mode 0, Command 0x9F (Read JEDEC ID)
signal "CS#" 1000000001
signal "SCLK" 0nnnnnnnn0
signal "MOSI" x========x "1,0,0,1,1,1,1,1"
signal "MISO" zzzzzzzzzz
mark 1 "CS# LOW, BIT 7 OUT"
mark 1.5 "SAMPLED" dash
mark 9 "CS# HIGH"
:::
```



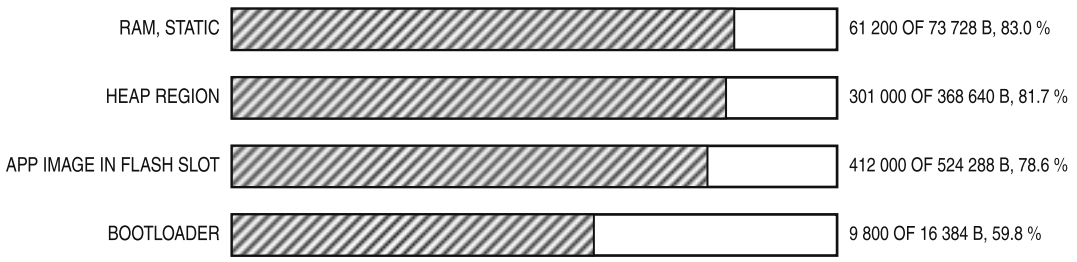
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Figure A-6. SPI Mode 0, Command 0x9F (Read JEDEC ID)

A.5.6 Budget gauges.

Used against capacity, one gauge a line, hatched to the used fraction, with the figures and the percentage at the end; hex is read as hex. `mark PERCENT "LABEL"` sets a limit across them all. A firmware image's memory:

```
:::budget Firmware 1.0 Memory
bar "RAM, static" 61200 73728 B
bar "Heap region" 301000 368640 B
bar "App image in flash slot" 412000 0x80000 B
bar "Bootloader" 9800 0x4000 B
:::
```



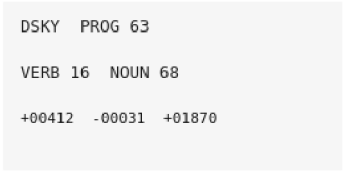
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Figure A-7. Firmware 1.0 Memory

A.5.7 Pictures and columns.

`![Caption](path){width=half}` takes an SVG, inlined so its lettering stays live, or a PNG; `width=third` and `gray=1` (a dark screen capture, printed inverted as ink on paper) are the options. `::::row` sets its contents side by side, columns split on a line `|||`, closed by `::::`; anything can go in a column.

```
![A Screen Capture](../kitchen-
sink/screen.png){width=half gray=1}
```



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Figure A-8. A Dark Screen Capture, Printed as Ink

A.6 MAKING A SHEET.

TURN A REPORT INTO A SHEET — **WRITER**

```
1  python sheet.py report.md --png
     -s section=N -s tab=XXX -s doc=XXX-N
     -s mission=... -s date=MM.DD.YYYY
2  read every page PNG; fix what crowds
3  add the figures: diagram, plot, budget,
     sequence, signals, timeline
4  equations: numbered, where-lists
5  cautions and checklists from the text
6  byline: DRAFT until checked
7  send the draft to the fact checker
8  apply corrections; name the checker
9  commit; send the PDF
```

For a sheet of your own, copy `template.md`: it has the front matter and one of each block, with comments that do not print. Later changes go in `:::changed` with `change:` set in the front matter; the basic date stays.

CAUTION

Chrome does not print into the page margins: the gutter numbers, edge tab and change bars are drawn inside the content box, so keep `main`'s padding when restyling. The faces come from Ubuntu's fonts-urw-base35 package (`/usr/share/fonts/opentype/urw-base35`). A block closes at the first line of exactly its own colons, so a block that holds blocks takes four: `::::changed` around an `:::equations`, like `::::row`. A row's own source cannot be shown inside a row.